

A Native Multi-Signal Model Router for LiteForge: Tiny Expert Panels, a RouteLLM Port, and an Honest Benchmark Study

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Abstract

We present the model-routing subsystem of LiteForge, an open-source LLM SDK. The router has two composable layers: a Layer-1 load balancer (deployments, strategies, health-aware cooldowns, fallbacks) and a Layer-2 content/quality selector. For Layer 2 we contribute (i) a native, dependency-light port of ROUTELLM’s matrix-factorization (MF) quality router, and (ii) a *panel* router: four independent tiny BERT experts (task type, difficulty, reasoning depth, context demand) plus structured codebase-context features, combined by a learned fusion matrix into capability groups. All components are small (each expert is a 4.4M parameter `bert-tiny`) and serve behind an OpenAI-compatible gateway. We then run an honest benchmark study on ROUTERBENCH. Our central finding is a cautionary one: classifiers trained on synthetic routing data, and even small encoders fine-tuned from scratch on real routing labels, transfer poorly to ROUTERBENCH’s cost-quality task, hovering near the random baseline. A lightweight logistic head over strong pretrained `bge-m3` embeddings is the only small router that meaningfully beats random, consistent with ROUTELLM’s own findings and with our MF design. We release the full training and evaluation harness.

1 Introduction

Serving large language models forces a cost-quality trade-off: the strongest models are many times more expensive than competent cheap ones, yet most queries do not need the strongest model. *Model routing* sends each query to an appropriate model. We describe a production router built natively into LiteForge [6] and, crucially, subject it to a rigorous external benchmark rather than reporting only in-distribution numbers.

Contributions.

1. A two-layer router: LiteLLM-style load balancing [2] plus a pluggable content/quality selector, reusing the SDK’s transport unchanged.
2. A native Rust port of ROUTELLM’s matrix-factorization router [5] (no transformer inference; a few matrix multiplies over a remote embedding).
3. A *panel* router: four independent tiny BERT experts and structured codebase-context features fused into capability groups.
4. An honest benchmark study on ROUTERBENCH [3]: zero-shot transfer of synthetic-trained routers, a from-scratch retrain, and an embedding-head retrain, with all code released.

2 Related Work

ROUTE_{LLM} [5] learns to route between a strong and weak model from preference data, with four routers (similarity-weighted, matrix factorization, a BERT classifier, and a causal-LLM classifier); they recommend matrix factorization and report up to 70% cost reduction at fixed MT-Bench quality. They evaluate with the call-performance threshold (CPT) and average performance gain recovered (APGR). ROUTER_{BENCH} [3] is the standard cost-quality benchmark: over 30k prompts with each of 11 LLMs’ correctness and cost plus oracle labels, evaluated with a cost-quality area metric. ROUTER_{EVAL} [4] frames routing as m -way classification at scale. Semantic routers [1] and LiteLLM’s auto-router route by embedding similarity to example utterances. Our panel differs by decomposing the decision into several independent small classifiers plus explicit structured features, then fusing them; this is closest in spirit to m -way classification routers but with an interpretable signal decomposition.

3 System Architecture

3.1 Two layers

A request carries a logical model name (or the alias `auto`). Layer 2 (a `ModelSelector`) returns a ranked list of model groups; Layer 1 load-balances within the top group and falls back down the ranking on failure. The selector trait lives in the Layer-1 module so the router can call it without a dependency cycle. Layer 1 mints a per-deployment configuration and reuses the SDK’s existing HTTP transport, so no request-path rewrite was required. Strategies are weighted-shuffle, round-robin, least-busy, and latency-based (an exponentially weighted latency average); failing deployments are cooled down after a configurable number of consecutive failures.

3.2 The panel router

Let a prompt (with any attached codebase context) be x . The panel comprises $K=4$ independent classifiers f_k , each a fine-tuned `bert-tiny` encoder (4.4M parameters) producing a probability vector $p_k = f_k(x) \in \Delta^{c_k}$ over its label set:

- task type ($c=9$: chitchat, qa, code, debug, refactor, math, reasoning, writing, data),
- difficulty ($c=3$: easy, medium, hard),
- reasoning depth ($c=2$: shallow, deep),
- context demand ($c=2$: low, high).

We also extract a structured feature vector $s(x) \in \mathbb{R}^5$ from the context: normalized token count, normalized file count, and binary has-code / has-diff / has-error flags. The fusion mapper is a learned linear map $W \in \mathbb{R}^{G \times d}$, $b \in \mathbb{R}^G$ over the concatenation $z = [p_1; p_2; p_3; p_4; s(x)] \in \mathbb{R}^d$, $d=21$, producing capability-group scores

$$\text{group}(x) = \arg \max_{g \in \mathcal{G}} \text{softmax}(Wz + b)_g, \quad \mathcal{G} = \{\text{chat, code, reasoning, long_context, general}\}.$$

The four experts run independently (and in parallel); the fusion is a single 21×5 matrix, so the whole panel is small and interpretable: each expert and feature contributes a named, inspectable signal.

3.3 Native RouteLLM MF port

We port ROUTE_{LLM}’s MF forward pass to dependency-light Rust. For a prompt embedding e (from `bge-m3`) and a learned per-model row r_m :

$$\text{pe} = W_{\text{proj}}^\top e + b_{\text{proj}}, \quad \ell_m = W_{\text{cls}}^\top (\widehat{r}_m \odot \text{pe}) + b_{\text{cls}}, \quad h(x) = \sigma(\ell_{\text{strong}} - \ell_{\text{weak}}),$$

where $\hat{\cdot}$ is L2 normalization, $\odot$ the Hadamard product, and $h \in (0, 1)$ a scalar “hardness”. There is no transformer inference: the embedding is fetched over HTTP and the rest is small linear algebra. Because MF is binary, we obtain N -way routing by bucketing h across the catalog’s ordered capability tiers.

3.4 Serving

Each Layer-2 selector can call a classifier served behind the LiteLLM gateway as an OpenAI-compatible model. We deploy two: **router-bert** (a single difficulty classifier) and **router-panel** (the full panel returning group scores plus the per-signal breakdown). The selector forwards the full message list so codebase context reaches the panel.

4 Training Methodology

Synthetic data. We generate a multi-signal dataset from coherent prompt archetypes that jointly set all four signal labels and attach synthetic code, diff, error, and file-list context. Each expert trains on prompt→its own label; the fusion trains on (expert one-hots + features)→a rule-derived capability group. On a held-out synthetic split every expert and the fusion exceed 99% accuracy, but this measures only in-distribution separability.

Retraining on real data. For the benchmark study we also train routers directly on ROUTERBENCH: (a) a **bert-mini** (11M) encoder fine-tuned from scratch to predict per-prompt model success, and (b) a logistic-regression head over frozen **bge-m3** embeddings, the lightweight embedding-router design that matches our MF selector. Both predict whether the weak model suffices; route-to-strong propensity is $1 - \hat{P}(\text{weak correct})$.

5 Experimental Setup

Benchmark. ROUTERBENCH (0-shot) [3]: 36,497 prompts drawn from MMLU, GSM8K, MBPP, HellaSwag, Winogrande, ARC, MT-Bench and others, each with 0/1 correctness and a cost for 11 LLMs and an oracle column. All quality and cost numbers are precomputed, so the study requires no LLM inference.

Cost-quality protocol. Following ROUTELLM, we use a binary strong/weak frame: weak = **mistral-8x7b-chat** (mean accuracy 0.548, cost $\$1.3 \times 10^{-4}$), strong = **gpt-4-1106-preview** (0.786, $\$3.3 \times 10^{-3}$). A router emits a route-to-strong propensity; sweeping the routed fraction traces a cost-quality curve between all-weak and all-strong. We report APGR (area between the router curve and the random line, normalized by the oracle) and the cost saved at 95% of strong-only quality, on a stratified 2,000-prompt test subsample (seed 13); retrained routers are trained on the disjoint remainder.

Intrinsic protocol. We derive provenance labels (MMLU/HellaSwag/ARC → qa; GSM8K → math; MBPP → code) and a difficulty proxy from which model tier first solves each prompt, then measure the panel experts’ accuracy on 2,027 real prompts (an out-of-distribution test of the synthetic-trained experts).

6 Results

6.1 Intrinsic out-of-distribution accuracy

Table 1 shows the synthetic-trained experts do not transfer to real prompts. Coarse task-type accuracy is 0.23 (three classes), and the difficulty experts sit near chance. The confusion (Figure 1) is informative: GSM8K word problems are read as “qa” rather than “math” (730/800), because the synthetic math archetypes were phrased as “prove/derive” statements; MMLU multiple-choice items are frequently read as “code”. The signal decomposition is sound, but the experts learned the surface form of the synthetic data.

Table 1: Intrinsic OOD accuracy on real ROUTERBENCH prompts (synthetic-trained).

Expert / signal	Accuracy	Macro-F1
Task type (coarse, 3-way)	0.230	0.198
Difficulty (panel)	0.326	0.264
Difficulty (router-bert)	0.235	0.139

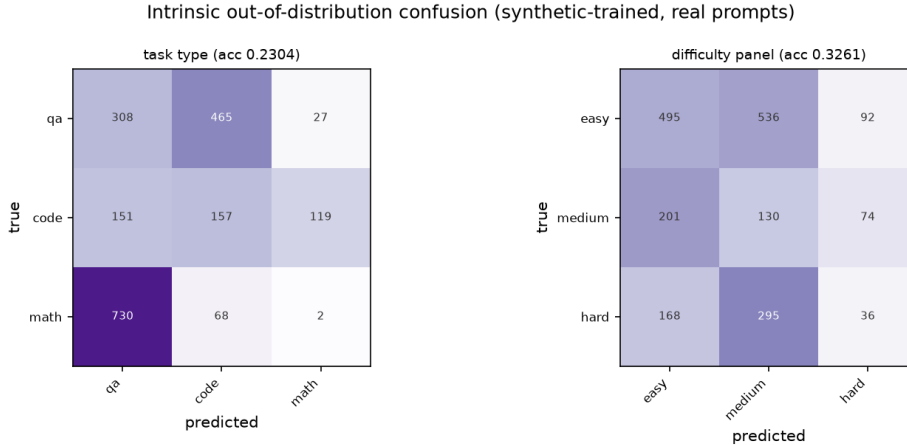


Figure 1: Out-of-distribution confusion for the synthetic-trained task-type and difficulty experts on real ROUTERBENCH prompts.

6.2 RouterBench cost-quality

Table 2 and Figure 2 give the headline result. Zero-shot, every synthetic-trained router is near the random baseline (APGR ≈ 0); the MF port is marginally positive (APGR 0.10) because it reasons over pretrained `bge-m3` embeddings rather than memorized surface form. Fine-tuning a small encoder (`bert-mini`) from scratch on real routing labels *fails to learn* (training loss plateaus at $\ln 2$; APGR -0.11). Only the logistic head over frozen `bge-m3` embeddings meaningfully beats random (APGR 0.308), recovering a substantial fraction of the oracle’s gain and saving 22.5% of cost at 95% of strong-only quality.

Optimization and deployment. A sweep over $\{\text{logistic, MLP}\} \times \{\text{binary weak-suffices, 3-class oracle-tier}\} \times \{\text{embedding, embedding+structural}\}$ selected by held-out APGR found logistic regression with structural features best (APGR 0.308); the MLP overfits at this data scale (APGR ≤ 0.12) and the 3-class target underperforms binary on the cost-quality curve. A companion task head over the same embeddings, trained on RouterBench provenance labels, classifies qa/code/math at 0.993 accuracy (versus 0.23 for the synthetic task expert), restoring

code-aware capability routing. This head is now LiteForge’s default router, served two ways: a native in-process `EmbeddingHeadSelector` (one embedding call plus a linear head) and a rebuilt numpy-only `router-panel` container behind the gateway. The learned quality axis plus cheap policy (triviality, code markers, context size) fuses into capability groups; ambiguous prompts default to a mid tier rather than the cheapest, since the RouterBench-trained head is unreliable on open-ended prompts outside its training distribution.

Table 2: ROUTERBENCH cost-quality (strong/weak frame; APGR: 0=random, 1=oracle).

Router	Training	APGR	Cost saved @95% strong
<code>router-bert</code> (difficulty)	synthetic	−0.358	7.5%
<code>panel</code> (difficulty signal)	synthetic	−0.033	10.4%
MF port	synthetic + bge-m3	+0.100	8.6%
<code>bert-mini</code> (from scratch)	RouterBench	−0.107	10.2%
embedding head (bge-m3, +struct) [default]	RouterBench	0.308	22.5%

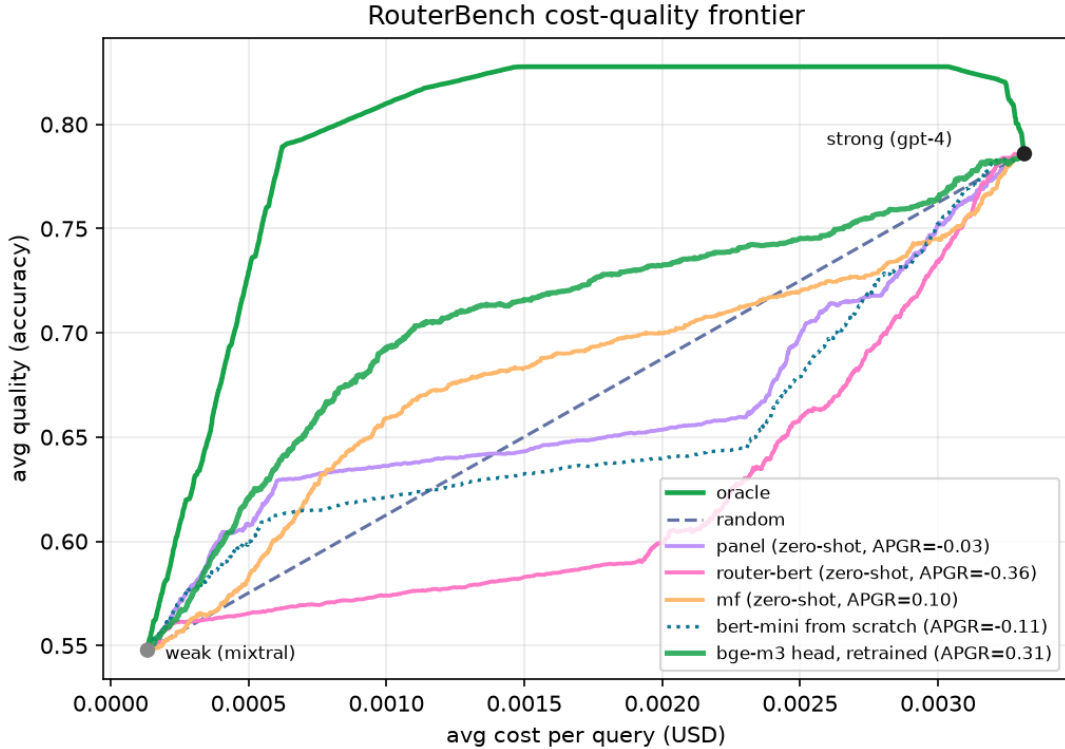


Figure 2: ROUTERBENCH cost-quality frontier. The oracle is the upper envelope; the dashed line is random mixing. Synthetic-trained routers sit near random; the embedding head trained on ROUTERBENCH climbs toward the oracle.

6.3 Live end-to-end check

Routing eight representative prompts through the deployed `router-panel` and the LiteLLM gateway behaves as designed: greetings and arithmetic route to the chat tier (`claude-haiku`), code and debugging to the code tier (`claude-sonnet`), proofs and system design to the reasoning tier (`claude-opus`); mean panel latency was 934 ms (one gateway round-trip).

7 Discussion and Limitations

The study is deliberately self-critical. (1) **Synthetic data does not transfer**: high in-distribution accuracy (99%+) coexists with near-random real-world routing, a reminder that router quality must be measured on real cost-quality data, not on the training distribution. (2) **Tiny from-scratch encoders underfit** the routing task; capacity and, more importantly, pretrained representations matter. (3) **Embeddings are the workhorse**: a trivial linear head over `bge-m3` is the only small router that beats random, echoing ROUTELLM’s finding that matrix factorization over embeddings is their best router, and validating our MF design choice. (4) Our panel emits abstract capability groups, so the zero-shot ROUTERBENCH mapping is approximate; the retrained heads route on real labels. (5) Costs are ROUTERBENCH’s API-list prices, which differ from a local gateway’s economics.

8 Conclusion

LiteForge ships a small, interpretable, native model router with a panel of tiny experts, a RouteLLM MF port, and load balancing, all behind an OpenAI-compatible gateway. Our benchmark study shows that the engineering is sound but that *representation*, not classifier head size, determines routing quality: synthetic training and from-scratch tiny encoders do not transfer, while a lightweight head over strong pretrained embeddings does. Future work: train the panel’s experts and fusion on real, embedding-based features; calibrate confidence; and route over a richer, capability-tagged model pool.

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